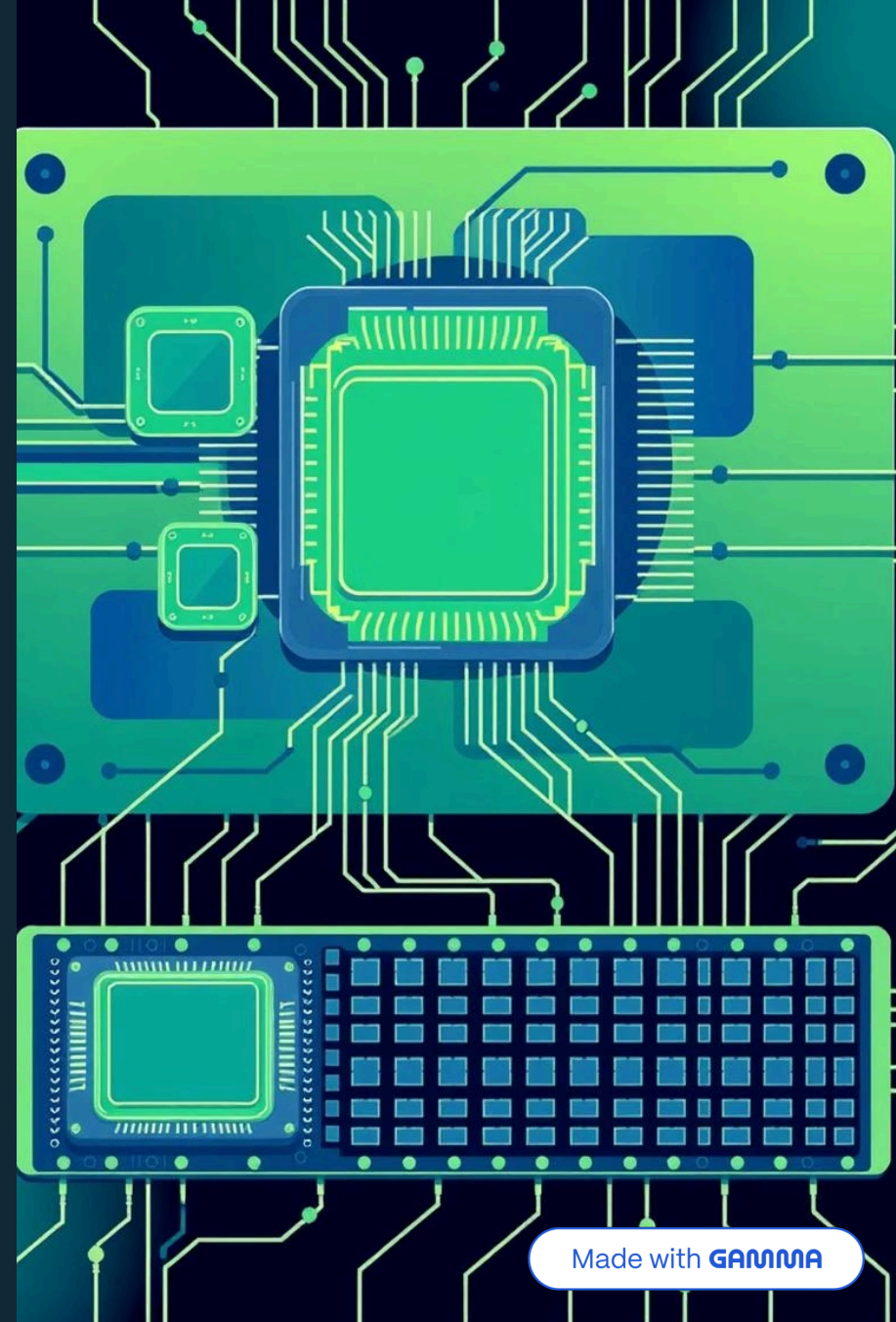
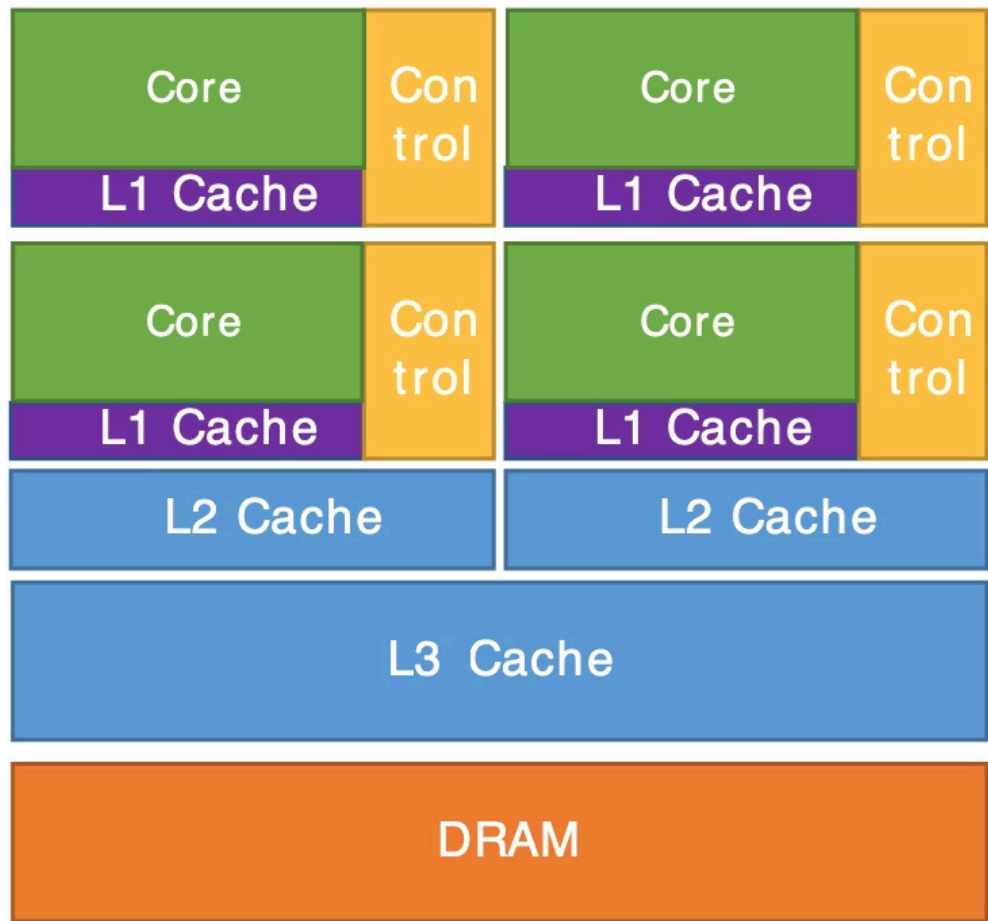


CPU vs GPU – The Big Picture

Understanding the fundamental differences between CPUs and GPUs is key to modern computing.

- **CPU:** Few *powerful* cores → best for sequential tasks.
- **GPU:** Hundreds/thousands of *smaller* cores → best for parallel tasks.
- Think: CPU = brainy sprinter. GPU = a crowd of steady workers.





CPU



GPU

Why Both Matter

CPU Strengths

- Great at decision-making, complex logic, sequential steps.
- Runs your OS, servers, apps.

GPU Strengths

- Excels when *the same operation* must be repeated across huge data.
- Example: multiplying millions of pixels or matrix elements.

Together: CPU is the “manager,” GPU is the “worker army.”

Problems That Map to GPUs

1

Data-Parallel Tasks

Same work on lots of pieces of data.

2

Image Filtering

Pixel by pixel processing.

3

Matrix Multiplication

Efficient handling of large mathematical operations.

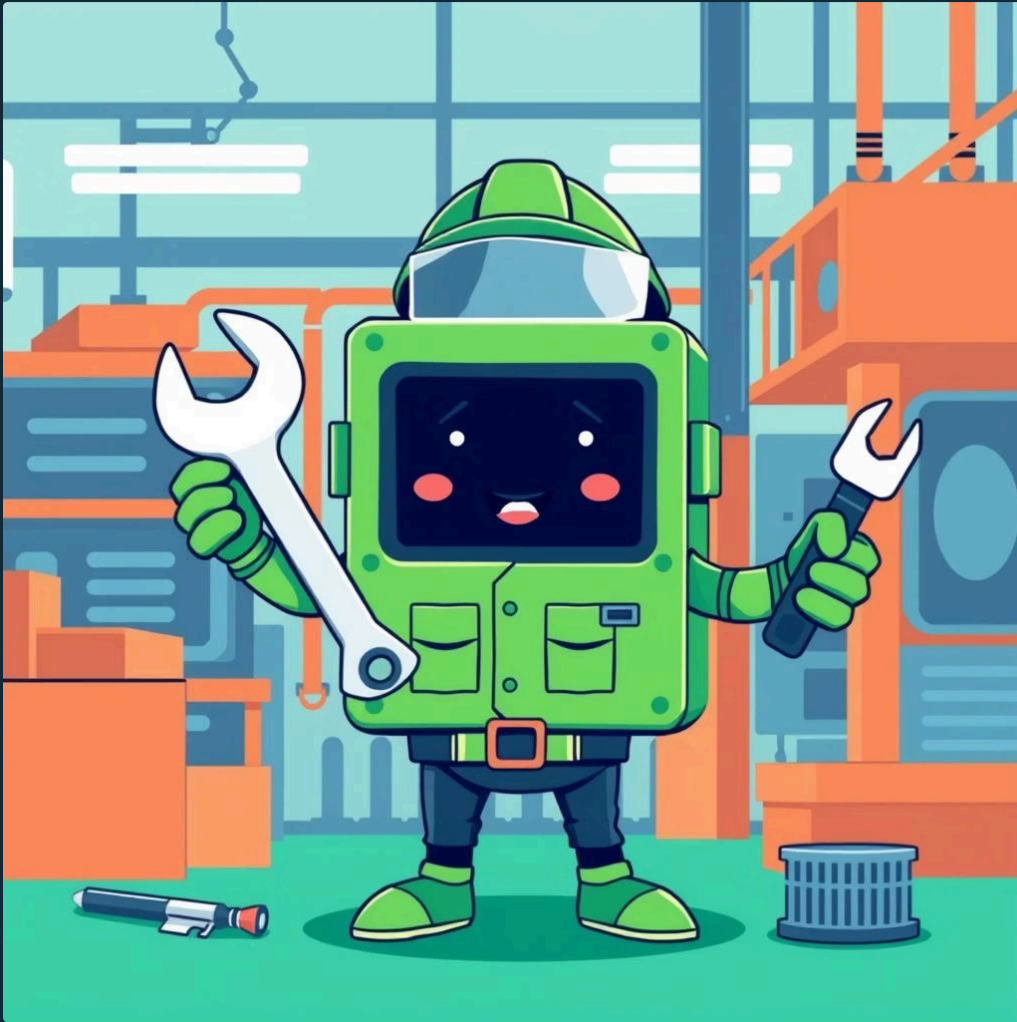
4

Machine Learning Training

Millions of weights updated at once.

CPU hands over the “big repetitive job,” GPU executes *in bulk*.

Analogy – Workers at a Factory



CPU (Swiss-Army Worker)

One person who can do many different things... but one at a time.

If a task can be broken into identical steps → GPU crushes it.



GPU (Assembly Line)

Hundreds of workers doing the *same simple job* in parallel.

Why It Matters

Modern computing = **scale**
and speed.

GPUs unlock:

- Faster ML model training.
- Real-time graphics and video processing.
- High-performance simulations.

